Name: Mohamed Wael Mohamed Allam

Aws Part:

In IAM:1) Creating Devops group

User groups (1) Info

A user group is a collection of IAM

Search

Group name
devops

2) Creating user1 and user2 and add them to devops group

Users (3) Info

An IAM user is an identity with long-term creder

Search

User name
mohamed-wael
user1
user2

3) Creating policy

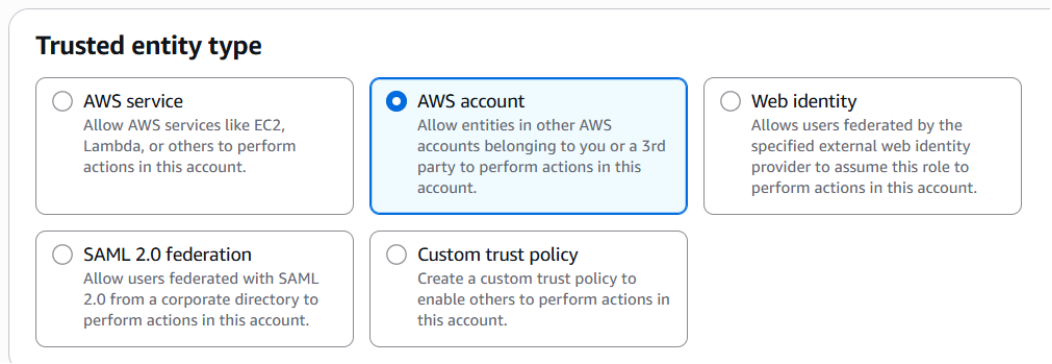
Policy editor

```
1  {
2    "Version": "2012-10-17",
3    "Statement": [
4      {
5        "Effect": "Allow",
6        "Action": [
7          "eks:*",
8          "ec2:*",
9          "iam:PassRole"
10       ],
11       "Resource": "*"
12     }
13   ]
14 }
```

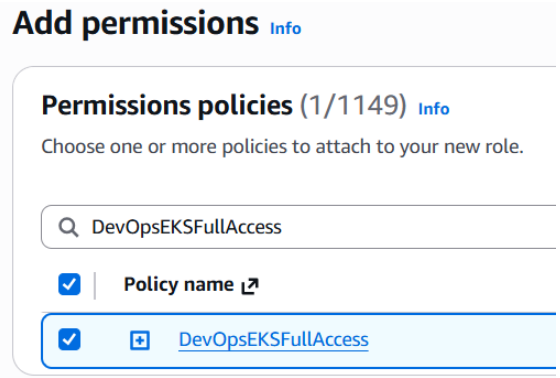
DevOpsEKSFullAccess

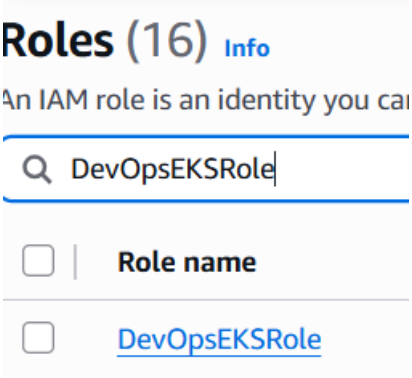
Policy name
DevOpsEKSFullAccess

4) Creating role



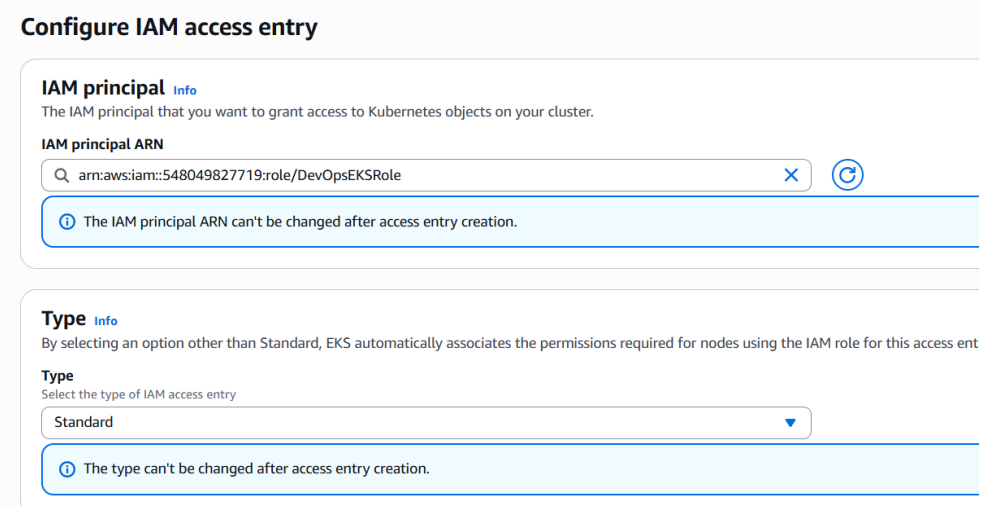
Attach our policy to it

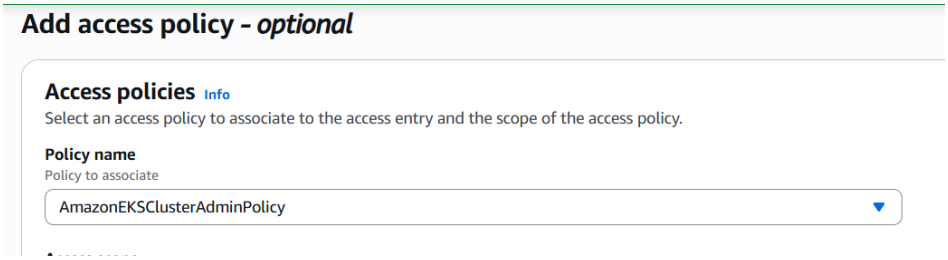




In EKS:

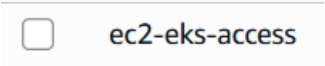
1) Creating Cluster





In EC2:

1) Creating EC2



Ansible Part:

1) Update system and check for kubectl

```
ubuntu@ip-10-0-0-11:~$ sudo ./aws/install
You can now run: /usr/local/bin/aws --version
ubuntu@ip-10-0-0-11:~$ aws --version
aws-cli/2.34.32 Python/3.14.4 Linux/6.17.0-1007-aws exe/x86_64.ubuntu.24
ubuntu@ip-10-0-0-11:~$ curl -LO "https://dl.k8s.io/release/$(curl -L -s https://dl.k8s.io/release/stable.txt)/bin/linux/amd64/kubectl"
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 55.8M  100 55.8M    0     0   130M      0 --:--:-- --:--:-- --:--:--  129M
ubuntu@ip-10-0-0-11:~$ chmod +x kubectl
ubuntu@ip-10-0-0-11:~$ sudo mv kubectl /usr/local/bin/
ubuntu@ip-10-0-0-11:~$ kubectl version --client
Client Version: v1.35.4
Kustomize Version: v5.7.1
```

2) Install Ansible

```
ubuntu@ip-10-0-0-11:~$ sudo apt install ansible -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

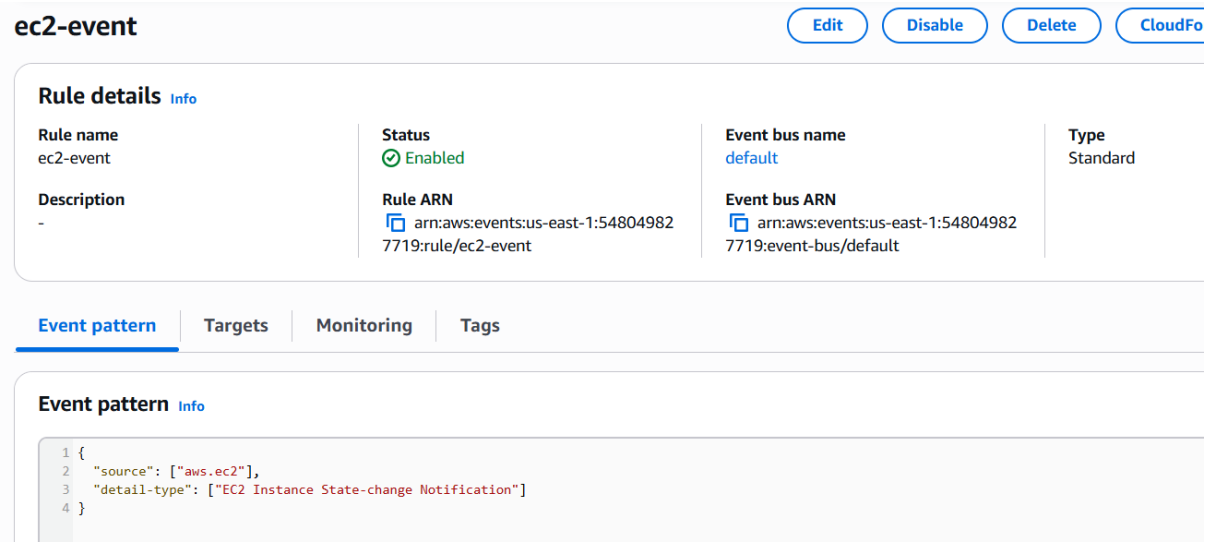
3) Check for kubectl

```
ubuntu@ip-10-0-0-11:~$ aws eks update-kubeconfig --region us-east-1 --name devops-cluster-test
Updated context arn:aws:eks:us-east-1:548049827719:cluster/devops-cluster-test in /home/ubuntu/.kube/config
ubuntu@ip-10-0-0-11:~$ kubectl get nodes
No resources found
```

4) deploy + service+ playbook

```
ubuntu@ip-10-0-0-11:~/ansible-eks$ vim deployment.yaml
ubuntu@ip-10-0-0-11:~/ansible-eks$ vim service.yaml
ubuntu@ip-10-0-0-11:~/ansible-eks$ ansible-playbook playbook.yml
[WARNING]: No inventory was parsed, only implicit localhost is available
[WARNING]: provided hosts list is empty, only localhost is available. Note that the implicit localhost does not match 'all'

PLAY [localhost] ***************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [localhost]

TASK [Apply deployment] ********************************************************************************************
changed: [localhost]

TASK [Apply service] ***********************************************************************************************
changed: [localhost]

TASK [Create deployment] *******************************************************************************************
changed: [localhost]

TASK [Create service] **********************************************************************************************
changed: [localhost]

PLAY RECAP *********************************************************************************************************
localhost                  : ok=5    changed=4    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

5) Event Bridge



Last: pushing